

Supplemental Materials

Supplemental Table 1. Robustness of phrase duration variability results to syllable-selection method. Phrase duration standard deviation (SD) was recalculated using multiple syllable-selection criteria. For each bird, syllables were selected within-animal and then averaged to produce one animal-level value per epoch. The primary Figure 3 analysis used the top 30% of syllables selected by pooled late-pre/post variability. ML indicates pooled complete and partial medial+lateral lesions. Δ SD indicates post-lesion SD minus late pre-lesion SD. Bootstrap confidence intervals and label-shuffle tests were calculated at the animal level.

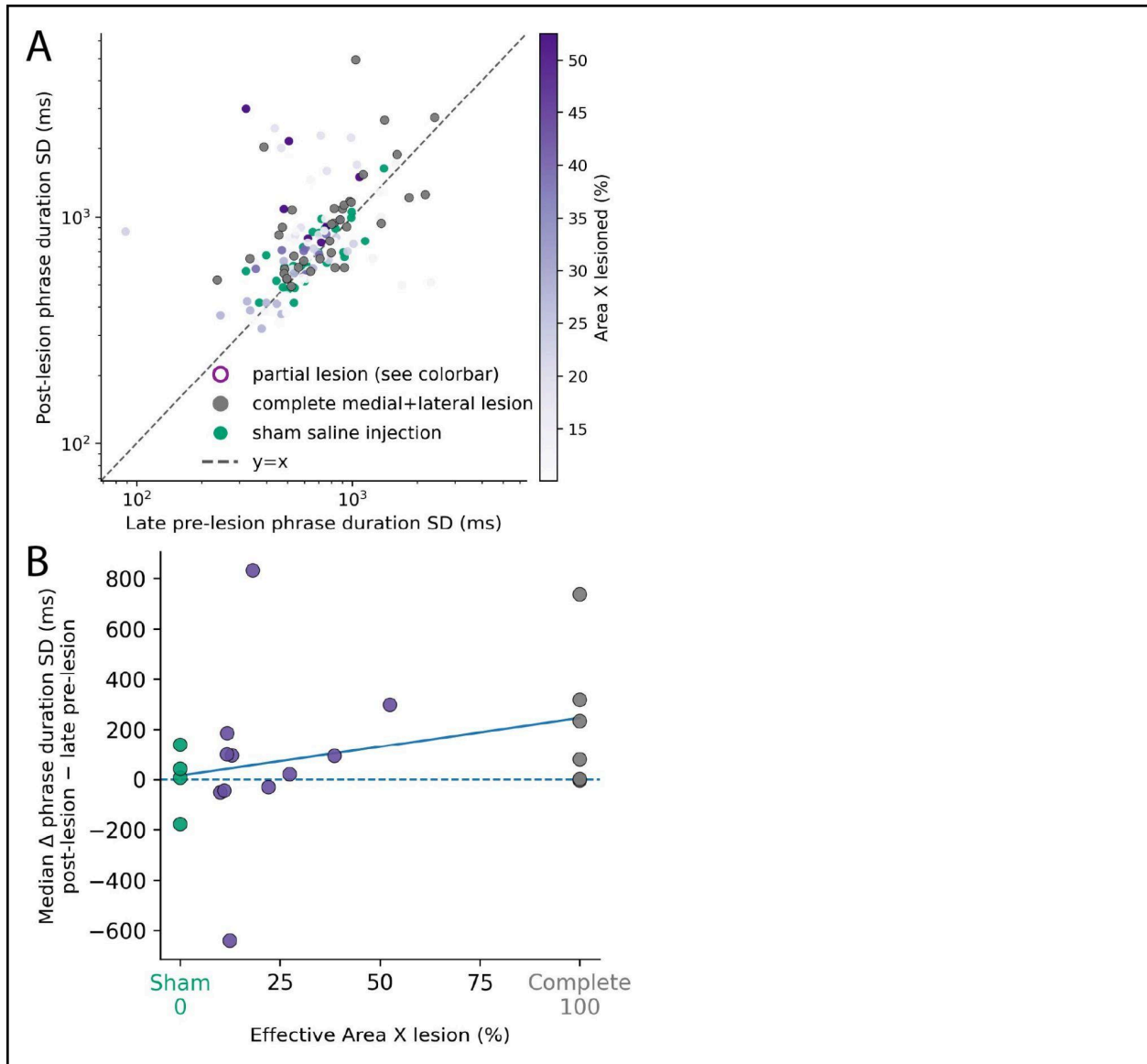
Selection method	Selection basis	ML median Δ SD (s)	ML post > pre p	ML sign-flip p	ML – lateral Δ SD (s)	95% bootstrap CI (s)	ML > lateral p	Significance
Post-selected top 20%	post-lesion selection	0.152	0.0300	0.0331	0.094	-0.090 to 1.215	0.1895	n.s.
Post-selected top 30%	post-lesion selection	0.109	0.0294	0.0331	0.064	-0.096 to 0.880	0.2372	n.s.
Post-selected top 40%	post-lesion selection	0.089	0.0292	0.0331	0.076	-0.042 to 0.728	0.0918	n.s.
Pooled pre+post-selected top 30%	pooled late-pre/post selection	0.109	0.0376	0.0331	0.186	-0.083 to 0.969	0.0338	*
Pre-selected top 30% validation	late pre-lesion selection	0.109	0.0834	0.1111	0.186	-0.046 to 0.641	0.0142	*
Held-out post-selected top 30% (day split)	held-out post-lesion day validation	0.286	0.0303	0.0350	0.238	-0.129 to 0.856	0.0494	*

Supplemental Table 2. Animal metadata and sample sizes for the primary Figure 3 phrase duration analysis. Each row corresponds to one bird included in the phrase duration SD analysis. Candidate syllables are syllable labels with sufficient late-pre and post-lesion phrase-duration data. Selected syllables are the top 30% of syllables within each bird selected by pooled late-pre/post phrase duration variability for the primary Figure 3 analysis. The current exported analysis tables contain phrase-duration observation counts, not independent song-bout counts; therefore phrase observations are reported here rather than song counts.

Bird ID	Treatment date	Lesion hit type	Lesion group	Recorded days pre	Recorded days post	Candidate syllables	Selected syllables	Late-pre phrase observations (selected)	Post phrase observations (selected)
USA5271	2024-03-07	sham saline injection	Sham	16	34	25	8	19,876	19,876
USA5283	2024-03-05	sham saline injection	Sham	25	34	28	9	24,840	24,840
USA5494	2025-03-17	sham saline injection	Sham	57	18	21	7	105,444	105,444
USA5506	2025-03-04	sham saline injection	Sham	19	24	18	6	111,924	111,924
R08	2025-05-22	Area X visible (single hit)	Lateral-only	33	17	21	7	19,480	19,480
R09	2025-05-23	Area X visible (single hit)	Lateral-only	29	7	19	6	1,836	1,836
R10	2025-05-16	Area X visible (single hit)	Lateral-only	22	12	13	4	5,916	5,916

USA53 36	2024-01-23	Area X visible (single hit)	Lateral-only	44	13	27	8	8,638	8,638
USA53 47	2024-01-20	Area X visible (single hit)	Lateral-only	35	26	38	12	34,942	34,942
USA54 83	2025-03-18	Area X visible (single hit)	Lateral-only	39	30	13	4	144,384	144,384
USA54 99	2025-02-20	Area X visible (single hit)	Lateral-only	34	22	23	7	151,632	151,632
USA55 10	2025-03-20	Area X visible (single hit)	Lateral-only	34	24	13	4	39,204	39,204
USA52 88	2024-04-09	Area X visible (medial+lateral hit)	Partial medial+lateral	36	14	27	8	84,312	84,312
USA53 25	2024-02-13	Area X visible (medial+lateral hit)	Partial medial+lateral	62	18	31	10	60,540	60,540
USA53 26	2024-02-20	Area X visible (medial+lateral hit)	Partial medial+lateral	32	18	30	9	19,532	19,532
USA53 37	2024-04-10	large lesion Area X not visible	Complete medial+lateral	37	12	28	9	87,792	87,792
USA53 71	2024-04-16	large lesion Area X not visible	Complete medial+lateral	34	22	29	9	73,824	73,824
USA54 43	2024-04-20	large lesion Area X not visible	Complete medial+lateral	9	29	22	7	72,560	72,560
USA54 68	2024-06-28	large lesion Area X not visible	Complete medial+lateral	30	15	12	4	87,616	87,616
USA55 05	2024-07-02	large lesion Area X not visible	Complete medial+lateral	39	3	12	4	1,216	1,216
USA55 09	2024-05-01	large lesion Area X not visible	Complete medial+lateral	20	17	14	4	99,312	99,312

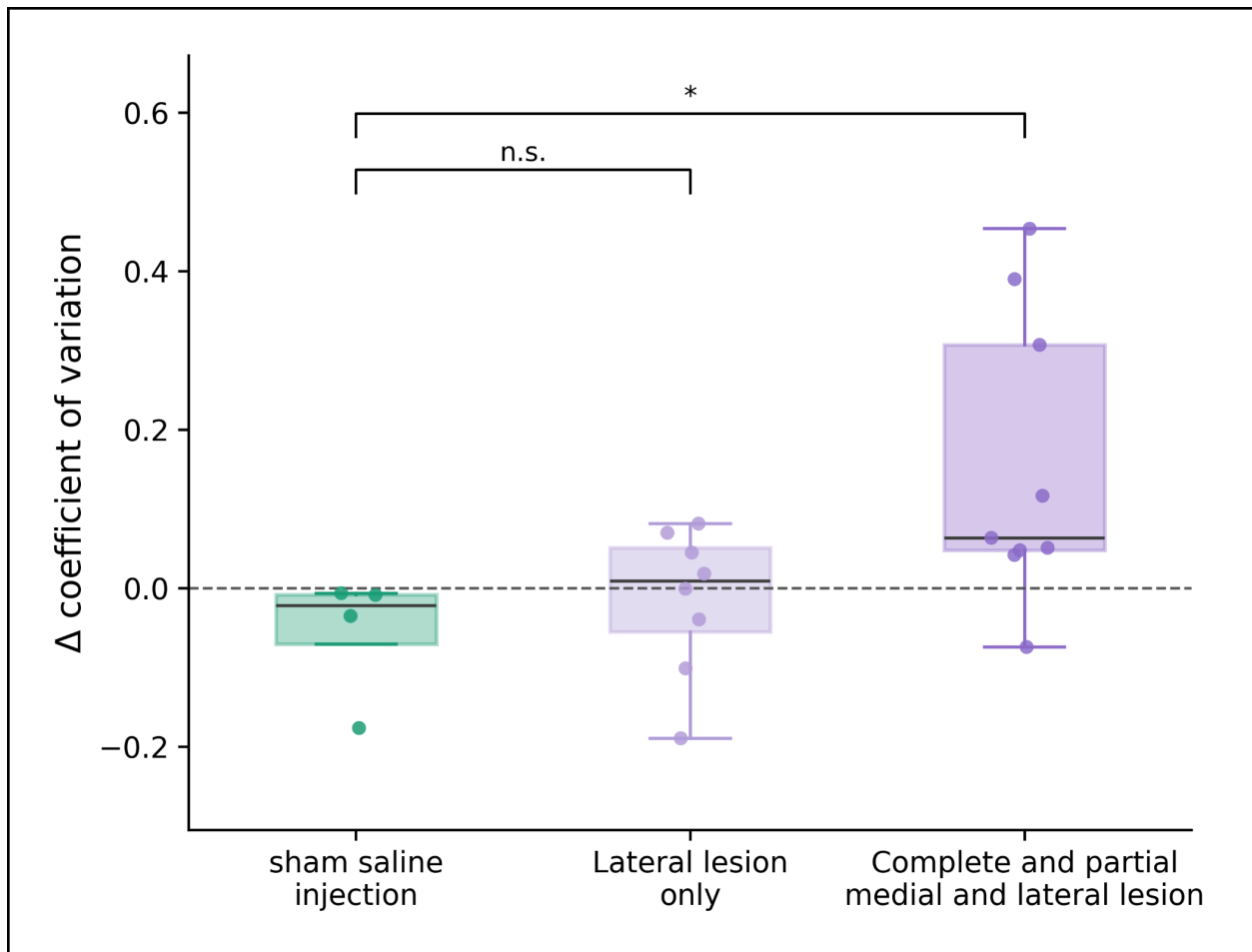
To test whether the phrase-duration variability effect depended on medial AFP involvement, we compared animal-level changes in phrase-duration standard deviation between medial+lateral and lateral-only lesion groups using bird-level label-shuffle permutation tests. Post-selected threshold analyses showed positive medial+lateral minus lateral-only differences across top 20%, 30%, and 40% syllable sets, although these direct contrasts did not reach significance. In contrast, validation analyses using less circular selection criteria showed significant anatomical specificity: medial+lateral lesions produced larger Δ SD than lateral-only lesions when syllables were selected using pooled pre/post variability (Δ = 185.9 ms, p = 0.0324) and when syllables were selected using late pre-lesion variability alone (Δ = 185.9 ms, p = 0.0129).



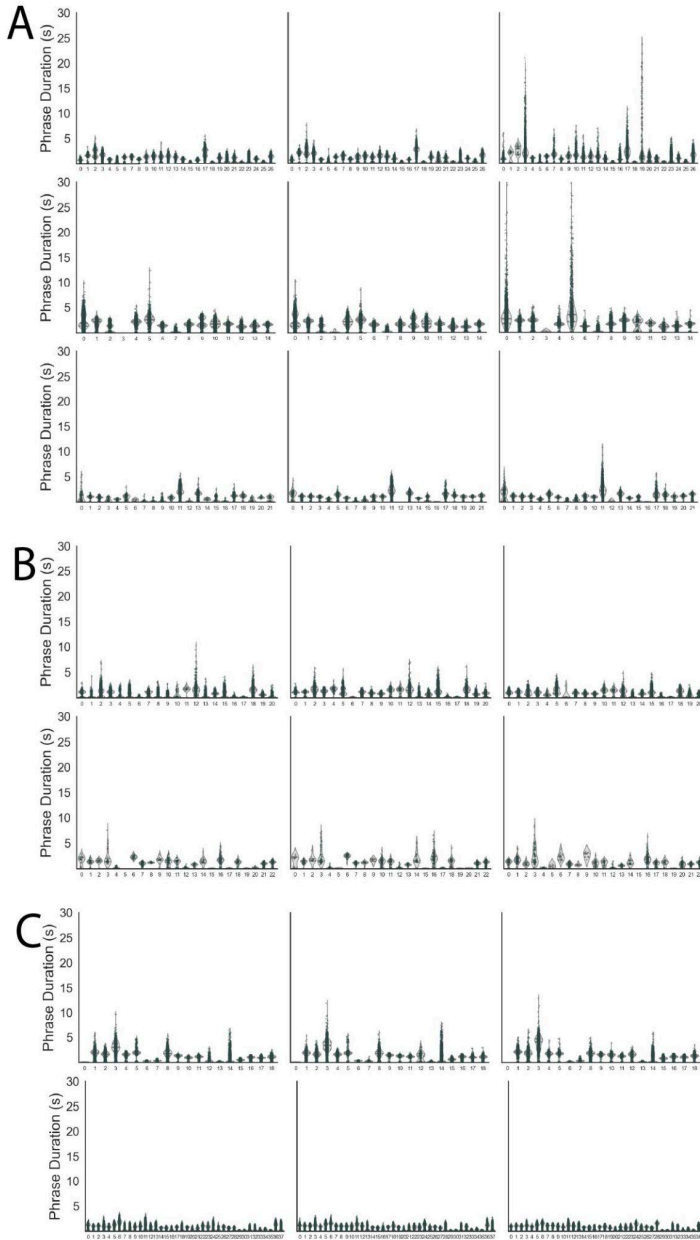
Supplemental Figure X. Lesion extent and phrase-duration variability correlation.

A) Late pre-lesion versus post-lesion phrase-duration standard deviation for the top 30% high-variability syllables. Each point represents one animal's syllable; points above the dashed $y=x$ line indicate increased post-lesion variability. Sham birds are teal, complete medial+lateral AFP lesions are grey, and partial lesions are purple, with darker shading indicating greater measured Area X lesion extent.

B) Animal-level relationship between effective Area X lesion extent and median change in phrase-duration standard deviation across the same selected syllables. Sham birds were assigned 0% lesion extent, partial lesions their measured lesion percentage, and complete medial+lateral AFP lesions 100%. Lesion extent was not significantly correlated with median Δ phrase-duration standard deviation (Spearman $\rho = 0.36$, $p = 0.10$, $n = 21$ animals).



Supplemental Figure X: Animal-level Δ phrase duration CV for pooled pre/post-selected top 30% syllables. Medial and lateral AFP lesions increased relative phrase-duration variability. Change in coefficient of variation (CV; phrase-duration SD / mean) from late pre-lesion to post-lesion song, calculated as the animal-level median Δ CV across selected syllables. Dashed line marks no change. Planned lesion-versus-sham label-shuffle tests were Holm-corrected across two comparisons. Lateral-only lesions did not differ from sham controls (Δ CV mean difference = 0.042, raw p = 0.229, Holm p = 0.229; n = 8 lateral, 4 sham). Medial+lateral lesions showed a significantly greater Δ CV than sham controls (Δ CV mean difference = 0.212, raw p = 0.0085, Holm p = 0.017; n = 9 medial+lateral, 4 sham). n.s., not significant; *Holm-corrected p < 0.05.



Supplemental Figure X. Representative phrase-duration distributions across lesion-hit categories. **A)** Representative birds with medial + lateral AFP lesions. **B)** Representative birds with lateral-only AFP lesions. **C)** Representative sham-control birds. Each row shows one bird, with phrase-duration distributions plotted separately for early pre-treatment, late pre-treatment, and post-treatment song. Within each plot, syllable labels are unique to that bird and are shown along the x-axis; violin plots with overlaid points show the distribution of phrase durations for each syllable label. Y-axes are fixed from 0–30 s across examples to facilitate comparison of long-duration phrase outliers. Medial + lateral lesion examples show the clearest post-treatment emergence of prolonged phrase durations, whereas lateral-only and sham-control examples show comparatively smaller or less consistent post-treatment changes.

